1.1 Limits of Functions Using Numerical and Graphical Techniques

Name _____ Period ____ Date _____

AP Calculus 1.1 · Student sheet

I can…

- use a table to investigate a limit.
- use a graph to find one-sided limits and decide whether a limit exists.
- describe the slope of a tangent line as a limit of the slopes of secant lines.

Vocabulary · fill in the missing word

Limit — $\lim f(x) = L$ as $x \to c$ means $f(x)$ gets as close to L as we like by _____ x close enough to c, $x \neq c$

One-sided limits — the _____ limit ($x \to c^-$) and the right-hand limit ($x \to c^+$)

Tangent line — the _____ position of the secant lines through P = (c, f(c)) and a point Q on the graph as Q moves closer to P

Part 1 · Closer and closer

$f(x) = (x^2 - 4)/(x - 2)$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
f(x)							

Part 2 · Notes and examples (fill in each note, then work the example)

Limit: f(x) gets as close to L as we like by taking x close enough to c, but x _____ c.

Example 1

Use a table to investigate the limit of $\frac{x^2 - 4}{x - 2}$ as $x \to 2$.

The value at c does not matter: the limit as $x \to c$ does not depend on f(_____); f can even be undefined there.

Example 2

$f(x) = x + 1$ for $x \neq 2$, and $f(2) = 5$. Find the limit of f(x) as $x \to 2$ and f(2).

One-sided limits: the limit as $x \to c$ is L if and only if both one-sided limits _____ L.

Example 3

Find the limit at c, if it exists.

a. $f(x) = 2x + 5$ if $x \leq 2$; $4x + 1$ if $x > 2$; $c = 2$

b. $f(x) = 2x + 1$ if $x \leq 0$; $2x$ if $x > 0$; $c = 0$

Example 4

Use a table (radians) to investigate the limit of $\frac{\sin x}{x}$ as $x \to 0$.

Slope of the tangent line: m_tan = the limit as $x \to c$ of $(f(x) - f(c))/(x - c)$, the limit of the slopes of _____ lines.

Example 5

$f(x) = 3x^2$. Find the slope of the secant line through (2, 12) and $(x, 3x^2)$, then the slope of the tangent line at (2, 12).

Part 3 · Partner practice

Partner 1

$f(x) = x^2$ if $x < 1$; $2x + 3$ if $x \geq 1$. Find the one-sided limits at 1 and the limit, if it exists.

Partner 2

(AP® Practice) The limits of f(x) as $x \to 3^-$ and as $x \to 3^+$ are both 5, and f(3) = 2. Which is true?

(A) The limit of f(x) as $x \to 3$ is 2. (B) The limit of f(x) as $x \to 3$ is 5.

(C) The limit of f(x) as $x \to 3$ does not exist. (D) f(3) = 5